

AMQUI, QC, Canada

WMO: 713860

Lat: 48.472N

Lon: 67.435W

Elev: 545

StdP: 14.41

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.2	-11.5	-22.4	1.6	-15.6	-18.0	2.1	-10.9	23.3	12.5	21.4	11.1	5.0	170	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	83.0	66.6	79.8	65.0	76.7	63.8	70.0	78.3	68.1	75.4	66.3	73.2	6.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	102.5	73.5	65.5	96.0	71.8	63.7	90.0	70.1	34.4	78.6	32.8	75.5	31.4	73.4	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.3	17.8	15.7	-25.4	88.5	5.8	2.7	-29.6	90.5	-33.0	92.1	-36.2	93.6	-40.4	95.5		
(5)			-25.0	73.4	4.8	2.4	-28.5	75.1	-31.3	76.5	-34.0	77.9	-37.4	79.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.4	9.4	11.8	21.6	35.2	48.9	58.5	64.6	63.0	54.7	42.7	30.8	17.9
	DBStd	21.00	11.99	11.33	11.01	7.40	7.32	6.67	5.07	5.41	7.25	7.39	8.72	10.63
	HDD50	5629	1259	1070	882	444	110	7	0	0	33	251	579	994
	HDD65	9841	1724	1490	1346	893	500	216	69	105	320	692	1027	1459
	CDD50	1394	0	0	0	2	77	261	454	402	172	24	2	0
	CDD65	131	0	0	0	0	2	20	58	42	9	0	0	0
	CDH74	1290	0	0	0	0	84	251	498	357	96	4	0	0
	CDH80	256	0	0	0	0	17	57	96	69	17	0	0	0
Wind	WSAvg	6.6	7.4	7.5	7.4	7.2	6.7	5.8	5.2	5.1	5.7	6.8	7.2	7.4
Precipitation	PrecAvg	41.8	3.1	2.8	3.0	2.7	3.3	3.6	4.3	3.8	3.8	3.9	3.8	3.7
	PrecMax	47.0	6.1	4.3	5.2	4.7	5.4	6.5	7.0	7.0	7.8	7.4	7.7	6.0
	PrecMin	33.9	0.6	1.1	1.6	0.9	1.2	1.0	1.0	0.8	1.3	1.1	0.8	1.8
	PrecStd	3.8	1.3	0.9	1.1	1.0	1.1	1.2	1.3	1.5	1.5	1.4	1.4	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.5	40.8	51.1	65.0	81.9	86.3	86.5	86.0	82.3	71.6	60.5	44.0
		MCWB	40.4	36.7	41.8	50.8	59.1	67.6	69.1	69.5	68.7	59.5	57.2	42.0
	2%	DB	36.4	36.6	44.8	56.7	75.3	80.5	82.7	81.6	76.4	63.6	51.2	38.9
		MCWB	34.9	34.2	37.5	44.8	57.6	63.2	66.8	67.3	64.7	56.7	47.8	37.0
	5%	DB	32.6	33.3	40.4	51.2	69.0	76.3	79.6	77.9	71.7	58.7	46.3	35.3
		MCWB	31.2	31.3	35.4	41.9	54.6	61.6	65.7	65.6	62.5	52.4	43.2	33.6
	10%	DB	27.5	29.7	37.1	46.9	63.7	72.1	76.3	74.6	67.4	54.4	42.2	32.9
		MCWB	25.8	27.6	33.0	39.7	51.8	60.2	64.5	63.9	60.2	50.0	39.6	31.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.6	38.0	44.1	53.2	64.7	70.9	73.1	71.8	70.8	61.9	56.9	42.5
		MCDB	41.3	39.1	47.9	60.9	74.1	81.6	80.0	81.5	78.9	67.4	60.0	43.4
	2%	WB	35.3	34.1	38.6	46.6	59.9	67.2	70.1	69.5	66.9	57.3	48.5	37.2
		MCDB	36.2	35.4	42.7	54.1	69.7	75.3	77.8	78.1	73.0	62.7	50.6	38.5
	5%	WB	31.0	30.9	36.0	43.2	56.5	64.6	68.2	67.5	64.2	53.8	43.3	34.0
		MCDB	32.6	32.9	40.1	49.5	66.0	72.3	75.5	74.5	70.3	58.0	45.3	35.1
	10%	WB	25.9	27.6	33.4	40.5	53.5	62.0	66.4	65.6	61.2	50.3	39.8	31.4
		MCDB	27.4	29.5	36.8	46.1	62.0	69.2	73.4	72.0	66.6	54.2	42.0	32.8
Mean Daily Temperature Range	5% DB	MDBR	17.0	18.6	19.2	17.0	20.3	20.2	18.8	19.1	18.7	14.6	11.5	13.3
		MCDBR	19.0	18.3	22.4	26.2	32.1	28.4	25.1	24.7	25.5	22.8	16.0	13.2
		MCWBR	18.5	15.8	16.5	15.9	16.4	13.5	12.0	12.6	15.0	15.4	13.5	12.7
	5% WB	MCDBR	19.0	15.7	18.7	22.6	25.9	21.9	19.8	20.6	22.0	19.9	14.8	13.5
		MCWBR	18.6	14.2	14.7	15.2	14.8	12.3	11.2	11.7	14.5	15.0	13.5	13.3
Clear-Sky Solar Irradiance	taub	0.255	0.271	0.301	0.361	0.355	0.373	0.400	0.379	0.356	0.327	0.292	0.274	
	taud	2.315	2.266	2.229	2.258	2.409	2.376	2.322	2.383	2.436	2.505	2.498	2.321	
	Ebn at Noon	261	282	290	281	287	282	272	273	269	260	248	236	
	Edh at Noon	22	30	36	40	36	37	38	35	30	24	19	20	
All-Sky Solar Radiation	RadAvg	392	673	1056	1384	1602	1733	1678	1453	1104	646	337	257	
	RadStd	46	75	75	118	123	122	177	98	83	56	44	24	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (22 neighbors)	N/A	N/A	+2.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page